Úl 1.2 $\langle , \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$: $\langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v}$

A reálná symetrická čtvercová matice $A = (a_{ij})$ řádu 2, $a_{11} > 0$ a $\det(A) > 0$

$\langle , \rangle$ je dr. součin na $\mathbb{R}^2$ ak:

1) $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$

2) $\langle \vec{u}, s\vec{v} \rangle = s \langle \vec{u}, \vec{v} \rangle$

3) $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$

4) $\langle \vec{u}, \vec{u} \rangle \geq 0$, a $\langle \vec{u}, \vec{u} \rangle = 0$ práve keď $\vec{u} = \vec{0}$

Budeme postupne dokazovať vlastnosti pre $\langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v}$:

1) $\vec{u}^T A \vec{v} = (\vec{v}^T A^T \vec{u}^T)^T = \vec{v}^T A^T \vec{u}$; A je symetrická, teda $A^T = A$ ✓

2) $\vec{u}^T A (s\vec{v}) = \vec{u}^T (sA\vec{v}) = s(\vec{u}^T A \vec{v})$ ✓

3) $\vec{u}^T A (\vec{v} + \vec{w}) = \vec{u}^T A \vec{v} + \vec{u}^T A \vec{w}$ ✓

4) $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$; $a_{11} > 0$; $\det(A) > 0$

$\langle \vec{u}, \vec{u} \rangle = \vec{u}^T A \vec{u} \geq 0$ ak $\vec{u} \neq \vec{0}$, $\vec{u} = \begin{pmatrix} b \\ c \end{pmatrix}$, $\vec{u}^T = (b \; c)$

Vyjadríme $\vec{u}^T A \vec{u}$ =

$$= (b \; c) \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} ba_{11} + ca_{12} \\ ba_{12} + ca_{22} \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = b(ba_{11} + ca_{12}) + c(ba_{12} + ca_{22}) =$$

$$= b^2 a_{11} + bca_{12} + bca_{12} + c^2 a_{22} \geq 0$$; doplníme do štvorca:

$$= a_{11}\left(b^2 + \frac{2bca_{12}}{a_{11}}\right) + c^2 a_{22} = a_{11}\left(b^2 + \frac{2bca_{12}}{a_{11}} + \left(\frac{a_{12}}{a_{11}} c\right)^2\right) + a_{22}c^2 - a_{11}\left(\frac{a_{12}}{a_{11}} c\right)^2 =$$

$$= a_{11}\left(b + \frac{a_{12}}{a_{11}} c\right)^2 + c^2\left(a_{22} - \frac{a_{12}^2}{a_{11}}\right) = a_{11}\left(b + \frac{a_{12}}{a_{11}} c\right)^2 + c^2\left(a_{22} - \frac{a_{12}^2}{a_{11}}\right)$$

≥ 0 ; viem, že $a_{11} > 0$, $c^2 \geq 0$, $\left(b + \frac{a_{12}}{a_{11}} c\right)^2 \geq 0$, $\left(a_{22} - \frac{a_{12}^2}{a_{11}}\right) =$

$$= \frac{a_{11}a_{22} - a_{12}^2}{a_{11}} = \frac{\det(A)}{a_{11}}$$ pre sym. maticu, a keďže $\det A > 0$ a $a_{11} > 0$,

tak $\frac{\det(A)}{a_{11}} > 0$ Pre celý výraz $a_{11}\left(b + \frac{a_{12}}{a_{11}} c\right)^2 + c^2\left(a_{22} - \frac{a_{12}^2}{a_{11}}\right) \geq 0$ Platí.

Ak $b, c = 0$, $\vec{u} = \vec{0}$, a potom $\langle \vec{u}, \vec{u} \rangle = 0$ taktiež platí

($a_{11} > 0$; $\left(b + \frac{a_{12}}{a_{11}} c\right)^2 > 0$ ak $b, c \neq 0$; $c^2 \geq 0$; $\left(a_{22} - \frac{a_{12}^2}{a_{11}}\right) > 0$)